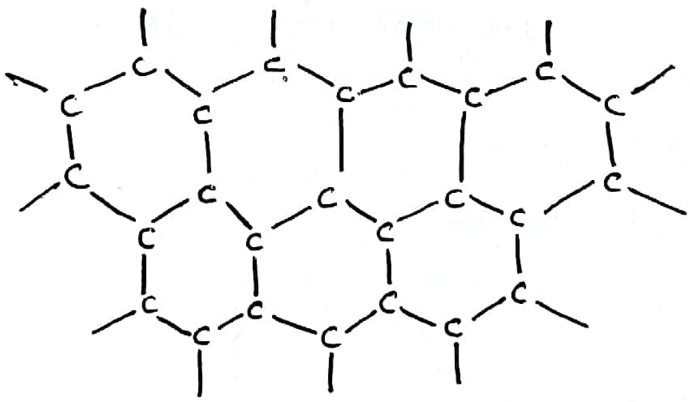
MITHBHASHI GUNATHILAKA

2

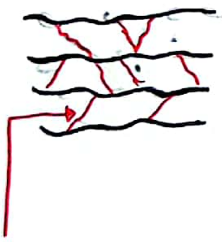
Table 14.1 Trends in Atomic, Physical, and Chemical Properties of the Period 2 Elements

Group: Element/At. No.:	1A(1) Lithium (Li) $Z = 3$	2A(2) Beryllium (Be) $Z = 4$	3A(13) Boron (B) $Z = 5$	4A(14) Carbon (C) $Z = 6$
Atomic Properties				
Condensed electron configuration; partial orbital diagram	[He] $2s^1$ [↑] [ ][ ][ ] 2s 2p	[He] $2s^2$ [↑↓] [ ][ ][ ] 2s 2p	[He] $2s^22p^1$ [↑↓] [↑][ ][ ] 2s 2p	[He] $2s^22p^2$ [↑↓] [↑][↑][ ] 2s 2p
Physical Properties				
Appearance				
Metallic character	Metal	Metal	Metalloid	Nonmetal
Hardness	Soft	Hard	Very hard	Graphite: soft Diamond: extremely hard
Melting point/ boiling point	Low mp for a metal	High mp	Extremely high mp	Extremely high mp

Graphene



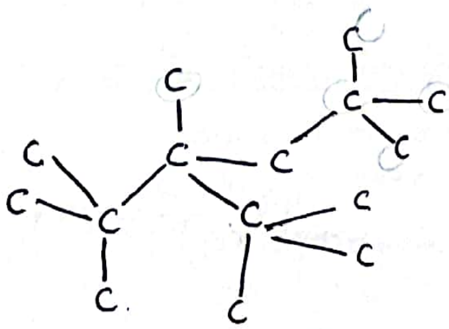
Graphite



weak Van der Waals forces

(These weak bonds cause for Graphite to be a soft solid.)

Diamond



⊛ Covalent bonded structure through out the solid.

⊛ Hardest material in the world.

do not have any other interactions between the layers.

2

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Atomic Properties				
Condensed electron configuration; partial orbital diagram	[He] $2s^1$ — 2s: ↑; 2p: empty	[He] $2s^2$ — 2s: ↑↓; 2p: empty	[He] $2s^22p^1$ — 2s: ↑↓; 2p: ↑	[He] $2s^22p^2$ — 2s: ↑↓; 2p: ↑ ↑
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Metallic character	Metal	Metal	Metalloid	Nonmetal
Hardness	$\boxed{\text{Soft}}$	$\boxed{\text{Hard}}$	Very hard	Graphite: soft Diamond: $\boxed{\text{extremely hard}}$
Melting point/boiling point	Low mp for a metal	High mp	Extremely high mp	Extremely high mp

Reducing agent → ඔක්සිහාරක

Table 14.1 Trends in Atomic, Physical, and Chemical Properties of the Period 2 Elements

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Chemical Properties				
General reactivity	$\boxed{\text{Reactive}}$	Low reactivity at room temperature	Low reactivity at room temperature	Low reactivity at room temperature; graphite more reactive
Bonding among atoms of element	Metallic	Metallic	Network covalent	Network covalent
Bonding with nonmetals	Ionic	Polar covalent	Polar covalent	Covalent (π bonds common)
Bonding with metals	Metallic	Metallic	Polar covalent	Polar covalent
Acid-base behavior of common oxide	Strongly basic	Amphoteric	Very weakly acidic	Very weakly acidic
Redox behavior (O.N.)	Strong reducing agent (+1)	Moderately strong reducing agent (+2)	Complex hydrides good reducing agents (+3, −3)	Every oxidation state from +4 to −4
Relevance/Uses of Element and Compounds	Li soaps for auto grease; thermonuclear bombs; high-voltage, low-weight batteries; treatment of bipolar disorders (Li_2CO_3)	Rocket nose cones; alloys for springs and gears; nuclear reactor parts; x-ray tubes	Cleaning agent (borax); eyewash, antiseptic (boric acid); armor (B_4C); borosilicate glass; plant nutrient	Graphite: lubricant (ලිහිසිතෙල්), structural fiber Diamond: jewelry, cutting tools, protective films Limestone ($CaCO_3$) Organic compounds: drugs, fuels, textiles, etc.

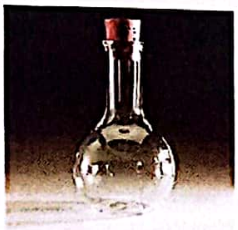
Graphite can be used to produce lubricant because the graphene layers graphite can slide with each other as the layers do not have any other interactions between the layers.

GUNATHILAKA

Nitrogen valance shell configuration is, $2s^2 2p^3$

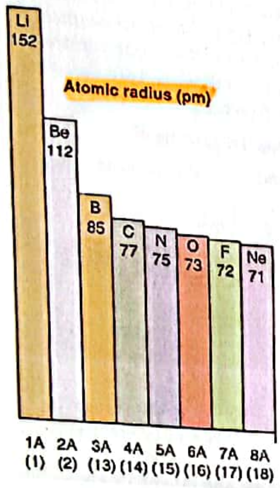
So in nature N always ~~remove~~ share $2p^3$ and get paired with anothe N atom to form N_2 molecule.

And also by sharing 3 electrons triple bond which is very strong and that makes N_2 less reactive

5A(15) Nitrogen (N) $Z = 7$	6A(16) Oxygen (O) $Z = 8$	7A(17) Fluorine (F) $Z = 9$	8A(18) Neon (Ne) $Z = 10$
[He] $2s^2 2p^3$ 2s: ↑↓ 2p: ↑ ↑ ↑	[He] $2s^2 2p^4$ 2s: ↑↓ 2p: ↑↓ ↑ ↑	[He] $2s^2 2p^5$ 2s: ↑↓ 2p: ↑↓ ↑↓ ↑	[He] $2s^2 2p^6$ 2s: ↑↓ 2p: ↑↓ ↑↓ ↑↓
		No sample available	
Nonmetal	Nonmetal	Nonmetal	Nonmetal
—	—	—	—
Very low mp and bp	Very low mp and bp	Very low mp and bp	Extremely low mp and bp

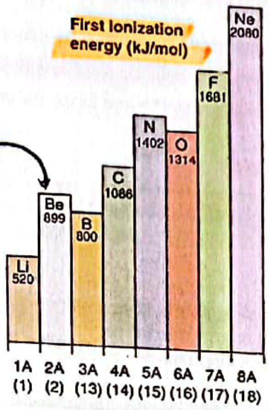
Chemical Properties	5A(15) Nitrogen (N) $Z = 7$	6A(16) Oxygen (O) $Z = 8$	7A(17) Fluorine (F) $Z = 9$	8A(18) Neon (Ne) $Z = 10$
General reactivity	Inactive at room temperature	Very reactive	Extremely reactive	Chemically inert
Bonding among atoms of element	Covalent N_2 molecules	Covalent O_2 (or O_3) molecules	Covalent F_2 molecules	None; separate atoms
Bonding with nonmetals	Covalent (π bonds common)	Covalent (π bonds common)	Covalent	None
Bonding with metals	Ionic/polar covalent; anions with active metals	Ionic	Ionic	None
Acid-base behavior of common oxide	Strongly acidic (NO_2)	—	Acidic	None
Redox behavior (O.N.)	Every oxidation state from +5 to −3	O_2 (and O_3) very strong oxidizing agents (−2)	Strongest oxidizing agent (−1)	None
	Component of proteins; ammonia for fertilizers, explosives; oxides involved in manufacturing and air pollution (smog, acid rain)	Final oxidizer in residential, industrial, and biological energy production	Manufacture of coatings (Teflon); glass etching (HF); refrigerants involved in ozone depletion (CFCs); dental protection (NaF, SnF_2)	Electrified gas in advertising signs

Because the $2s^2 2p^5$ configuration always tries to gain a electron to obtain the stable configuration of $2s^2 2p^6$ of Ne.

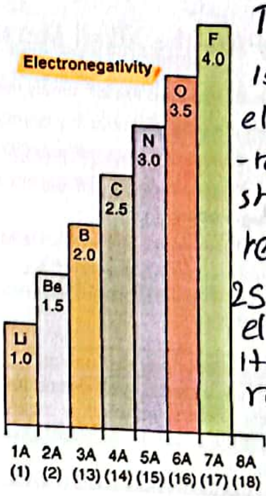


Li has the highest radius because Li has a single electron in its outermost shell, thus it has a small effective nuclear charge and a larger covalent radius of aprox.

$1s^2 2s^2$

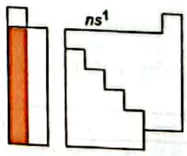


It is harder to remove an electron from Be than Li because the $2s^2$ electron configuration is more stable than the $2s^1$ electron configuration of Li.



To obtain the $1s^2$ stabilized electron configuration Li atoms struggle to remove the $2s^1$ valence electron. So, it is electron repulsive rather than attractive

14.3 GROUP 1A(1): THE ALKALI METALS.



GROUP 1A(1)

- The first group of elements in the periodic table form alkaline (basic) solutions when their oxides dissolved in water.
- All the elements in the group-lithium (Li), sodium (Na), potassium (K), rubidium (Rb), cesium (Cs), and rare, radioactive francium (Fr)*are very reactive metals.

Why Are the Alkali Metals Soft, Low Melting, and Lightweight?

- Unlike most metals, the alkali metals are soft
  - Na has the consistency of cold butter,
  - K can be squeezed like clay.
- The alkali metals have lower melting and boiling points than any other group of metals.
  - Except for Li, they all melt below 100°C
  - Cs melts a few degrees above room temperature
- They have lower densities than most metals
  - Li floats on lightweight household oil.

Lithium floating in oil floating on water.

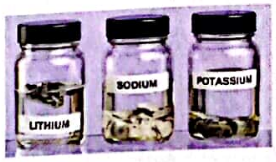
(*) These metals are named as Alkali metals for the alkaline (basic) nature of their oxides and for the basic solutions the elements forms in water.

Answer → This is mainly because of their atomic size, the largest in their respective periods, and to the ns^1 valence electron configuration. Because the single valence electron is relatively far from the nucleus, there is only a weak attraction between the delocalized electrons and the metal-ion cores in the crystal structure. This means the alkali metal crystal structure can be easily deformed or broken down, which results in a soft consistency and low melting point

Why Are the Alkali Metals So Reactive?

- The alkali metals are extremely reactive elements.
- They are powerful reducing agents and, thus, always occur in nature as 1+ cations rather than as free metals.
  - The alkali metals reduce the halogens and form ionic solids in reactions that release large quantities of heat.
  - They reduce the hydrogen in water, reacting vigorously (Rb and Cs explosively) to form H_2 and a metal hydroxide solution.
  - They reduce O_2 in the air, and thus tarnish rapidly.

ex-: $Na_{(s)} + \frac{1}{2}Cl_{2(g)} \rightarrow NaCl_{(s)}$



Because of this reactivity, Na and K are usually kept under mineral oil in the lab and Rb and Cs are handled with gloves under an inert argon atmosphere.

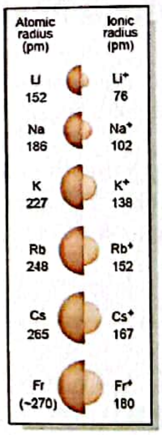
ns^1 configuration is the basis for their physical properties

1. *Low heat of atomization* (ΔH°_{atom}).
Consistent with the alkali metals' low melting and boiling points, their weak metallic bonding leads to low values for ΔH°_{atom} (the energy needed to convert the solid into individual gaseous atoms), which decrease down the group:

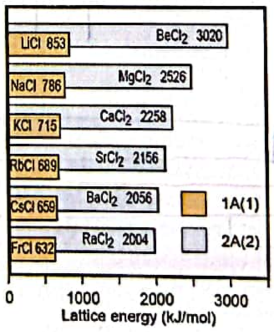
$$E(s) \longrightarrow E(g) \qquad \Delta H_{atom}\ (Li > Na > K > Rb > Cs)$$

ns^1 configuration is the basis for their physical properties

2. *Low IE and small ionic radius.*
- Each alkali metal has the largest size and the lowest IE in its period. A great decrease in size occurs when the outer electron is lost.
  - Volume of the Li ion is less than 13% that of the Li atom! (Because by forming Li^+ a whole energy level loses.)
  - Group 1A(1) ions are small spheres with considerable charge density.



3. High lattice energy.
- When Group 1A(1) salts crystallize, large amounts of energy are released because the small cations lie close to the anions.
- For a given anion, the trend in lattice energy is the inverse of the trend in cation size
  - As the cation becomes larger, the lattice energy becomes smaller.

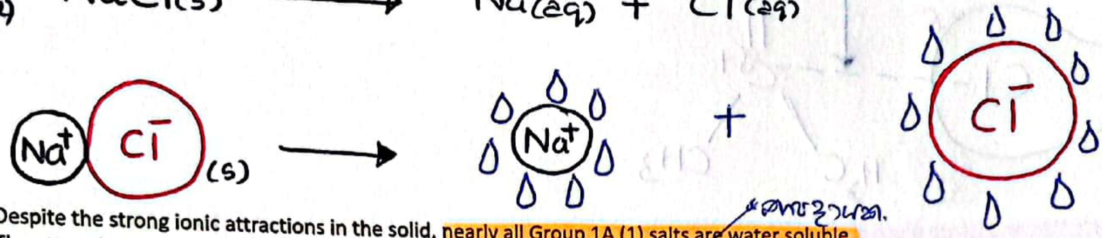


When the cation becomes larger, it is harder to keep closer to the anion. So, the anion. Then the stability of the latice obtain a lower value

$$\text{Latice energy} \propto \frac{1}{\text{size of the cation.}}$$

δ - H_2O

$H_2O_{(l)} + NaCl_{(s)} \longrightarrow Na^+_{(aq)} + Cl^-_{(aq)}$



Entropy the measure of system's thermal energy per unit temperature that is unavilable for doing useful work. Because work is obtained from ordered molecular motion, the amount of entropy is also a measure of the molecular disorder, or randomness of a system.

- Despite the strong ionic attractions in the solid, nearly all Group 1A (1) salts are water soluble.
- The attraction between the ions and water molecules creates a highly exothermic heat of hydration, and a large increase in entropy occurs when ions in the organized crystal become dispersed and hydrated in solution; together, these factors outweigh the high lattice energy.

The magnitude of the hydration energy *decreases* as ionic size increases:

$$E^+(g) \longrightarrow E^+(aq) \qquad \Delta H = -\Delta H_{hydr}\ (Li^+ > Na^+ > K^+ > Rb^+ > Cs^+)$$

$$\Delta E_H \propto \frac{1}{\text{ionic size}}$$

- Interestingly, the smaller ions attract water molecules strongly enough to form larger hydrated ions. This size trend has a major effect on the function of nerves, kidneys, and cell membranes because the sizes of $Na^+(aq)$ and $K^+(aq)$, the most common cations in cell fluids, influence their movement in and out of cells.

There are two reasons for this.

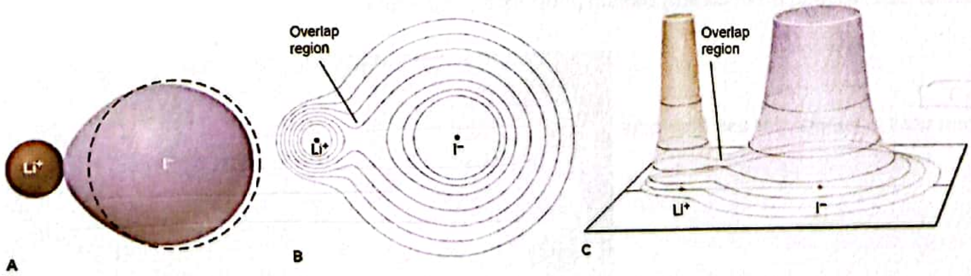
① Attraction between the ions and the water molecules.

The Anomalous Behavior of Lithium

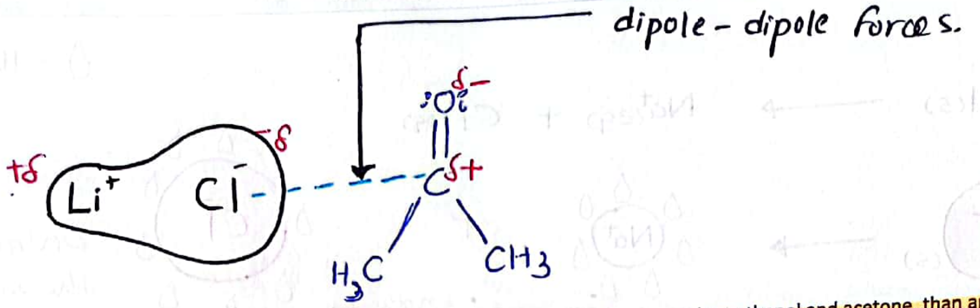
- It is the only member that forms a simple oxide and nitride, Li_2O and Li_3N on reaction with O_2 and N_2 in air.
- Only Li forms molecular compounds with hydrocarbon groups from organic halides:

$$2Li(s) + CH_3CH_2Cl(g) \longrightarrow CH_3CH_2Li(s) + LiCl(s)$$

- Because of its small size, Li has a relatively high charge density.
  - Therefore, it can deform nearby electron clouds to a much greater extent than the other 1A ions can, which increases orbital overlap and gives many lithium salts significant covalent character



A B C



- Thus, LiCl, LiBr, and LiI are much more soluble in polar organic solvents, such as ethanol and acetone, than are the halides of Na and K, because the lithium halide dipole interacts with these solvents through dipole-dipole forces.
- The small, highly positive Li makes dissociation of Li salts into ions more difficult in water; thus, the fluoride, carbonate, hydroxide, and phosphate of Li are much less soluble in water than those of Na and K

~~Because the Li^+ is so small;~~

Important Compounds

1. Lithium chloride and lithium bromide, LiCl and LiBr.
   Because the Li ion is so small, Li salts have a high affinity for H_2O and yet a positive heat of solution. Thus, they are used in dehumidifiers and air-cooling units.

2. Lithium carbonate, Li_2CO_3
   Used to make porcelain enamels and toughened glasses and as a drug in the treatment of bipolar disorders.

3. Sodium chloride, NaCl
   Millions of tons used in the industrial production of Na, NaOH, Na_2CO_3/$NaHCO_3$, Na_2SO_4, and HCl or purified for use as table salt.

4. Sodium carbonate and sodium hydrogen carbonate, Na_2CO_3 and $NaHCO_3$.
   Carbonate used as an industrial base and to make glass. Hydrogen carbonate, which releases CO_2 at low temperatures (50°C to 100C), used in baking powder and in fire extinguishers.

5. Sodium hydroxide, NaOH.
   Most important industrial base; used to make bleach, sodium phosphates, and alcohols.

6. Potassium nitrate, KNO_3.
   Powerful oxidizing agent used in gunpowder and fireworks

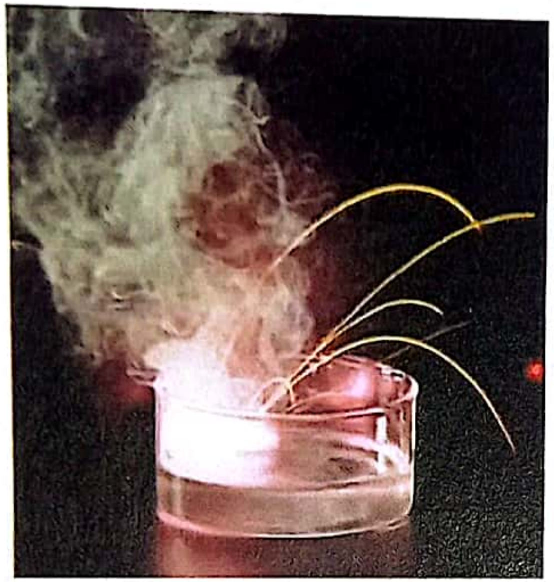
Important Reactions

The reducing power of the alkali metals (E) is shown in reactions 1 to 4. Some industrial applications of Group 1A(1) compounds are shown in reactions 5 to 7.

1. The alkali metals reduce H in H_2O from the +1 to the 0 oxidation state:

$$2E(s) + 2H_2O(l) \longrightarrow 2E^+(aq) + 2OH^-(aq) + H_2(g)$$

The reaction becomes more vigorous down the group (*see photo*).



Potassium reacting with water

2. The alkali metals reduce oxygen, but the product depends on the metal. Li forms the oxide, Li_2O; Na forms the peroxide, Na_2O_2; K, Rb, and Cs form the superoxide, EO_2:

$$4Li(s) + O_2(g) \longrightarrow 2Li_2O(s)$$

$$K(s) + O_2(g) \longrightarrow KO_2(s)$$

In emergency breathing units, KO_2 reacts with H_2O and CO_2 in exhaled air to release O_2.

3. The alkali metals reduce hydrogen to form ionic (saltlike) hydrides:

$$2E(s) + H_2(g) \longrightarrow 2EH(s)$$

NaH is an industrial base and reducing agent that is used to prepare other reducing agents, such as $NaBH_4$.

4. The alkali metals reduce halogens to form ionic halides:

$$2E(s) + X_2 \longrightarrow 2EX(s) \qquad (X = F, Cl, Br, I)$$

5. Sodium chloride is the most important alkali metal halide.

(a) In the Downs process for the production of sodium metal (*Section 22.4*), reaction 4 is reversed by supplying electricity to molten NaCl:

$$2NaCl(l) \xrightarrow{\text{electricity}} 2Na(l) + Cl_2(g)$$

(b) In the chlor-alkali process (*Section 22.5*), NaCl(*aq*) is electrolyzed to form several key industrial chemicals:

$$2NaCl(aq) + 2H_2O(l) \longrightarrow 2NaOH(aq) + H_2(g) + Cl_2(g)$$

(c) In its reaction with sulfuric acid, NaCl forms two major products:

$$2NaCl(s) + H_2SO_4(aq) \longrightarrow Na_2SO_4(aq) + 2HCl(g)$$

Sodium sulfate is important in the paper industry; HCl is essential in steel, plastics, textile, and food production.

6. Sodium hydroxide is used in the formation of bleaching solutions:

$$2NaOH(aq) + Cl_2(g) \longrightarrow NaClO(aq) + NaCl(aq) + H_2O(l)$$

7. In an ion-exchange process (*Chemical Connections, p. 542*), water is "softened" by removal of dissolved hard-water cations, which displace Na^+ from a resin:

$$M^{2+}(aq) + Na_2Z(s) \longrightarrow MZ(s) + 2Na^+(aq)$$

$$(M = Mg, Ca; Z = \text{resin})$$